

Empirical Research in Finance

Combined Course Submission · Group A

Humboldt-Universität zu Berlin
School of Business and Economics
Summer Semester 2026

PART I · GROUP SUBMISSION · EMPIRICAL ASSIGNMENT

Firm Valuation and Future Stock Returns

Evidence on the Value Effect and the Brexit Shock
among German Listed Firms

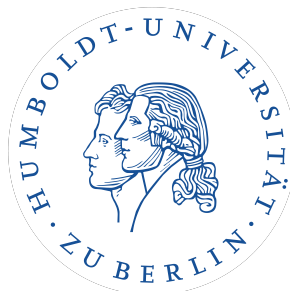
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Course	Reference	Instructor	Submission
Empirical Research in Finance	701143	Dr. Beatriz Mariano	15 September 2026



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PART II · INDIVIDUAL SUBMISSION · RESEARCH PROPOSAL

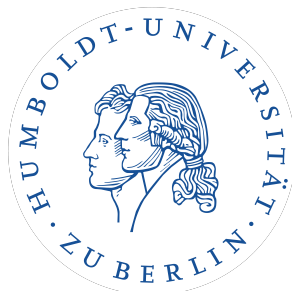
Weather Shocks and Corporate Innovation

R&D Investment and Precautionary Cash Holdings
after Hurricane Helene

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Background

Hurricane Helene made landfall in Florida's Big Bend region on 26 September 2024 and then moved inland. It caused about \$78.7 billion in damage across Georgia, South Carolina, North Carolina, Tennessee and Virginia. Different from many hurricanes, most of the destruction was not on the coast but inland, in counties where very few households have flood insurance, so most families and local firms had to cover the losses themselves. Because some counties were hit very hard while areas close by were almost untouched, Helene is a good case to compare firms that were directly exposed with firms that were not.

Dessaint and Matray (2017) find that US firms in counties hit by a hurricane hold more cash afterwards, even when their business did not really get worse. They explain this by salience: after managers experience a disaster, the risk feels more real and they become more careful. This proposal uses the same idea but looks at R&D instead. R&D is long-term and risky, and it is hard to finance with debt or to stop and restart, so it is likely to be cut first if managers become more risk-averse. The research question is: does direct exposure to Hurricane Helene cause firms to reduce R&D investment relative to unexposed firms, and is this decline accompanied by a precautionary increase in cash holdings?

Expected Outcome

My main hypothesis (H1) is that firms directly exposed to Helene cut their R&D intensity after September 2024 compared to unexposed firms, and at the same time hold more cash.

I expect this for two reasons. First, R&D is hard to keep after a shock. It only pays off in the long run, it is risky, and banks do not accept it as collateral, so it is difficult to finance with debt. It is also easier to postpone than repairing a damaged building. Cash is the opposite. It can be used for anything if another disaster happens, like repairs or to cover lower sales. So managers in hit areas probably prefer to keep cash as protection instead of spending on R&D.

This is close to what Dessaint and Matray (2017) find. They show that even firms that were only near a hurricane, not directly hit, increase cash by about one percentage point of assets. They also talk a lot more about hurricane risk in their annual reports after the shock. But the effect becomes smaller when firms have seen hurricanes before, and the extra cash is much larger than the actual expected losses. This suggests managers overreact because the risk feels very close, and investors value this extra cash less. For Helene I expect the same kind of reaction: R&D goes down while cash goes up. If only R&D fell, it could just mean sales were lower. But if cash rises at the same time, it points more to precaution.

Data

I will use a quarterly firm panel from Q1 2022 to Q4 2025. I use quarterly data because Helene happened at the end of September 2024, so yearly data would mix months before and after the

storm. The period gives about 11 quarters before Helene and 5 quarters after. This is needed to check that treated and control firms moved in a similar way before the shock, and to see if the effect after lasts or fades. I start in 2022 to avoid the COVID years, when R&D and cash were very unusual. I stop at the end of 2025 to have enough time after but not pick up much later shocks.

Firm accounting data come from Compustat Quarterly, accessed through WRDS. Compustat collects the quarterly reports of all US listed firms, so it gives me R&D, cash, assets and the other variables for every quarter. The main variable is R&D intensity, measured as $xrdq$ divided by atq . $xrdq$ is R&D spending in the quarter and atq is total assets in the quarter. I scale by assets and not by sales because sales itself can fall after a hurricane when shops close or supply chains break. That would move the ratio even if R&D did not change. I use the same denominator as for cash, which is $cheq$ divided by atq , so the two results are easy to compare. When $xrdq$ is missing I set it to zero, because Compustat leaves it empty when firms have no real R&D to report. I also control separately for firms that never report R&D. I drop financials (SIC 6000–6999) and utilities (SIC 4900–4999) and winsorize all ratios at 1% and 99%, because a few very small firms can have extreme values.

Treatment comes from FEMA Disaster Declarations for Helene and the NHC report on the storm path. FEMA is the official list that says which counties got federal help after Helene. A firm is treated if its headquarters county got such a declaration. Headquarters county comes from Compustat header data ($addzip$) and is checked with 10-Ks. All other US mainland firms are control. $Post$ is 1 from Q4 2024 on. Because quarterly R&D can be noisy, I will also check yearly R&D intensity (xrd/at) as a robustness test.

Methodology

I use difference-in-differences to compare treated and control firms before and after Helene. The main regression is:

$$Y_{it} = a_i + \ell_t + b \cdot \text{Treated}_i \cdot \text{Post}_t + g \cdot X_{it} + e_{it}$$

Y_{it} is first R&D intensity ($xrdq/atq$) and then cash holdings ($cheq/atq$). I run the same regression twice, once for each outcome. a_i are firm fixed effects. They control for everything about a firm that does not change over time, like industry or location. ℓ_t are quarter fixed effects. They control for shocks that hit all firms in the same quarter. X_{it} are controls: size, leverage, profitability and Tobin's Q. Standard errors are clustered by firm, because the same firm is observed many times.

Treated_i is 1 if the firm is in a FEMA county for Helene, and 0 if not. Post_t is 1 from Q4 2024 on, and 0 before. I drop Q3 2024 because Helene happened at the very end of September, so that quarter is neither fully before nor after. The coefficient b is what I care about. It shows how treated firms changed after Helene compared to control firms.

For H1 I expect $b < 0$ when Y is R&D, and $b > 0$ when Y is cash. If both happen together, it supports the idea that firms cut R&D to hold more cash as protection. This only works if both

groups moved in a similar way before Helene. I will check this with a plot of average R&D and cash for both groups, and by replacing Post with separate quarter dummies to see if there was any difference before.

Extensions and Limitations

One limitation is that headquarters county does not always show where a firm really operates. A firm can be headquartered outside the FEMA area but have factories or shops inside it, or the other way round. In that case I put firms in the wrong group. This kind of mistake normally makes the effect look smaller than it really is, so my b would go toward zero. A second problem is that cutting R&D automatically leaves more cash in the firm. So the two results alone cannot fully prove that managers acted out of precaution.

As an extension I will test if the effect is stronger for firms that had problems financing before Helene. I use a triple interaction $\text{Treated}_i \cdot \text{Post}_t \cdot \text{Constrained}_i$, where Constrained_i is 1 for firms with high financing constraints in 2023 and 0 if not. I measure constraints with firm size and leverage before the shock. If constrained firms cut R&D more and build up more cash than others, this supports the self-insurance idea, because these firms cannot easily borrow from banks after a shock.

Reference

Dessaint, O. and Matray, A. (2017). Do managers overreact to salient risks? Evidence from hurricane strikes. *Journal of Financial Economics*, 126(1), 97–121.
<https://doi.org/10.1016/j.jfineco.2017.01.002>